Addendum

Clarification on partial R^2 reporting suggestion

2025-08-31

In the letter of concern I stated that the "relevant quantity is the **partial $\mathbf{R^2}$** for the exposure conditional on baseline (and prespecified covariates) in an ANCOVA model" and suggested reporting "partial R^2 of exposure conditional on baseline (and prespecified covariates)". While this is appropriate in OLS ANCOVA, a clarification should be made.

In simple OLS settings, total/adjusted/partial R^2 can summarize fit. However, this does not generalize cleanly to more advanced models commonly used in this area. There is no single, standard R^2 across GLMs, mixed-effects models, or hurdle/two-part models, and available analogues are model-dependent and not directly comparable.

The study's linear models violate standard OLS assumptions (nonlinearity, heteroskedasticity, non-normal residuals). Therefore, a non-OLS specification should be used, and a standard R^2 and partial R^2 are not applicable.

The recommendation would be to report the coefficient (β) with 95% CI, and state effects on the outcome scale rather than as proportions of residual variance. For example: *Each 10 mg/dL higher ApoB is associated with a +0.25 percentage-point change in plaque progression.*